

JAYANTH DASAMANTHARAO

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PROFESSIONAL EXPERIENCE

Applied AI Engineer, EliteUS Software Solutions

Aug 2024 - Present | Edison, NJ

- Designed and deployed NLP-driven production systems leveraging text embeddings, semantic similarity, retrieval pipelines to extract structured insights from large unstructured datasets, serving 12,500+ users.
- Designed and productionized scalable ML microservices and REST APIs for NLP models, enabling secure, versioned, high-availability inference in AWS/Azure environments.
- Built feature extraction pipelines using embeddings and clustering techniques to identify patterns, group similar documents, and surface actionable signals from unstructured text data.
- Built distributed data pipelines for document ingestion, hybrid semantic search, reranking, and vector retrieval, indexing 10,000+ documents with low-latency fault-tolerant serving.

Data Science Intern, Harvest Software Solutions

June 2023 - Sept 2023 | Jacksonville, Florida

- Engineered scalable ML pipelines and monitoring systems for large advertising datasets, increasing campaign visibility from 15% to 45% while ensuring reliability and performance.
- Built NLP-driven classification models and APIs to extract sentiment and performance signals from text-based advertising data, improving campaign efficiency by 50%.
- Designed SQL/Python data processing and evaluation pipelines for feature engineering, model training, and validation, enabling rapid experimentation and improving engagement quality by 40%.

Application Development Associate, Accenture

June 2021 - Aug 2022 | Hyderabad, India

- Developed and optimized Python-based ETL and data processing pipelines using hash maps, sets, indexed lookups to efficiently clean, deduplicate, and transform large datasets, reducing production issues by 50%.
- Implemented SQL workflows and joins with partitioning, indexing, and window functions, enabling efficient aggregation and analysis over millions of records in AWS Athena and Redshift.
- Applied batching, caching, and algorithmic optimizations for data validation and pipeline execution, improving system performance and reliability by 25%.

TECHNICAL SKILLS

Programming: Python, R, SQL. **Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, FastAPI.
Machine Learning : Model Fine-tuning, evaluation and experimentation, deployment, Feature Engineering.
NLP & LLMs: NLP , text embeddings, semantic similarity, clustering, classification, feature extraction, RAG.
APIs & Services: REST APIs, API versioning, authentication, developer tooling, ETL, distributed processing.
Cloud & MLOps: PostgreSQL, Data Warehousing, AWS (Bedrock, Athena, Redshift), GCP, Azure, Vectordb.

EDUCATION

Rutgers University, Master of Science in Data Science

Sept 2022 - May 2024 | New Brunswick, NJ

Personal Website: <https://jayanthdasamantharao.github.io/>

Andhra University B.Tech in Electrical and Electronics Engineering

2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Enterprise NLP Retrieval & Insight Platform (Production): Built large-scale text ingestion and embedding pipelines to process enterprise documents, enabling semantic search, clustering, and structured insight extraction using RAG and evaluation-driven iteration.

Self-Driving Cars (Computer Vision): Built autonomous driving cars with vision pipeline using CNNs, YOLO, Faster R-CNN, achieving 95%+ object detection accuracy.

Bayesian Classification: Implemented Bayesian logistic regression in PyMC3 to predict diabetes status, incorporating informative priors, achieving 0.85 F1-score.

For additional projects and code samples: <https://github.com/JayanthDasamantharao>